

Dropout#

<https://pytorch.org/docs/stable/generated/torch.nn.Dropout.html>

Description:

During training, randomly zeroes some of the elements of the input tensor with probability p . The zeroed elements are chosen independently for each forward call and are sampled from a Bernoulli distribution. Each channel will be zeroed out independently on every forward call. This has proven to be an effective technique for regularization and preventing the co-adaptation of neurons as described in the paper Improving neural networks by preventing co-adaptation of feature detectors . Furthermore, the outputs are scaled by a factor of $\frac{1}{1-p}$ during training. This means that during evaluation the module simply computes an identity function.

Function Signature

```
class torch.nn.Dropout(p=0.5, inplace=False)[source]#
```

Parameters

Shape:

```
forward(input)[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> m = nn.Dropout(p=0.2)
>>> input = torch.randn(20, 16)
>>> output = m(input)
```

Detailed Documentation

Documentation

During training, randomly zeroes some of the elements of the input tensor with probability p .

The zeroed elements are chosen independently for each forward call and are sampled from a Bernoulli distribution.

Each channel will be zeroed out independently on every forward call.

This has proven to be an effective technique for regularization and preventing the co-adaptation of neurons as described in the paper Improving neural networks by preventing co-adaptation of feature detectors .

Furthermore, the outputs are scaled by a factor of $\frac{1}{1-p}$ during training. This means that during evaluation the module simply computes an identity function.

p (float) – probability of an element to be zeroed. Default: 0.5

`inplace` (bool) – If set to True, will do this operation in-place. Default: False

Input: $(*)$. Input can be of any shape

Output: $(*)$. Output is of the same shape as input

Runs the forward pass.
